v.0.0.4

최병권

# AI와 대화하기

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# 1

# 기획의도

Part 1

# 기획 의도

요즘 가장 유행하고 있는 Chat-GPT 영감을 받아서 Google Cloud에서 머신 러닝을 진행하였고, 일반 사람들과 다르지 않게 대화를 주고 받을 수 있게 작성해보았다. Google Cloud Platfrom(GCP)에서 Intents와 Entities를 추가하여 진행하였다. (intent, Entity는 대화를 주고 받기 위해 로봇 두뇌에 대답 리스트를 집어넣는 과정)

# 2

## 구성요소

- java 17
- Spring Boot
- Postgres

- Vue.js
- javascript
- Nuxt

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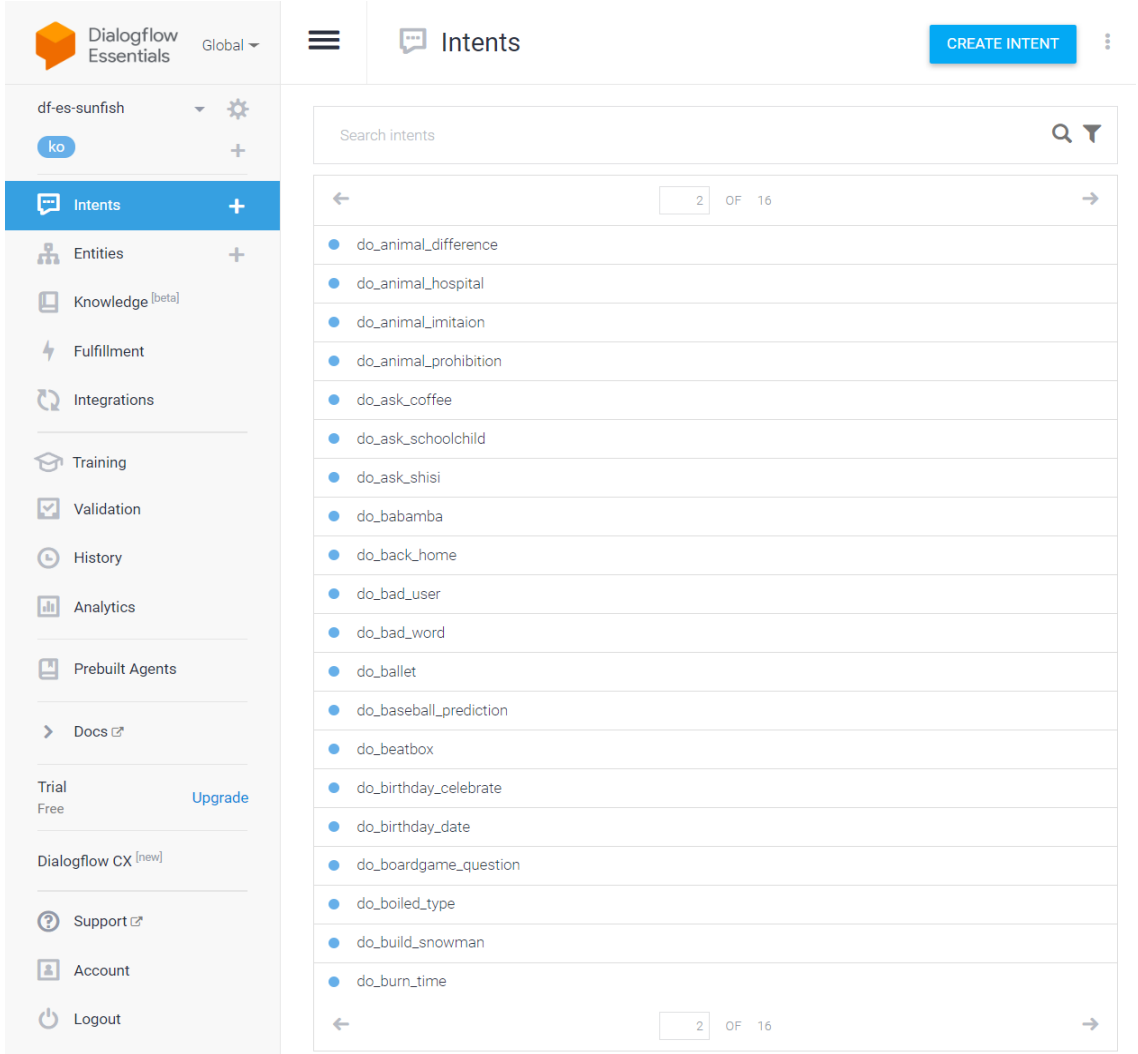
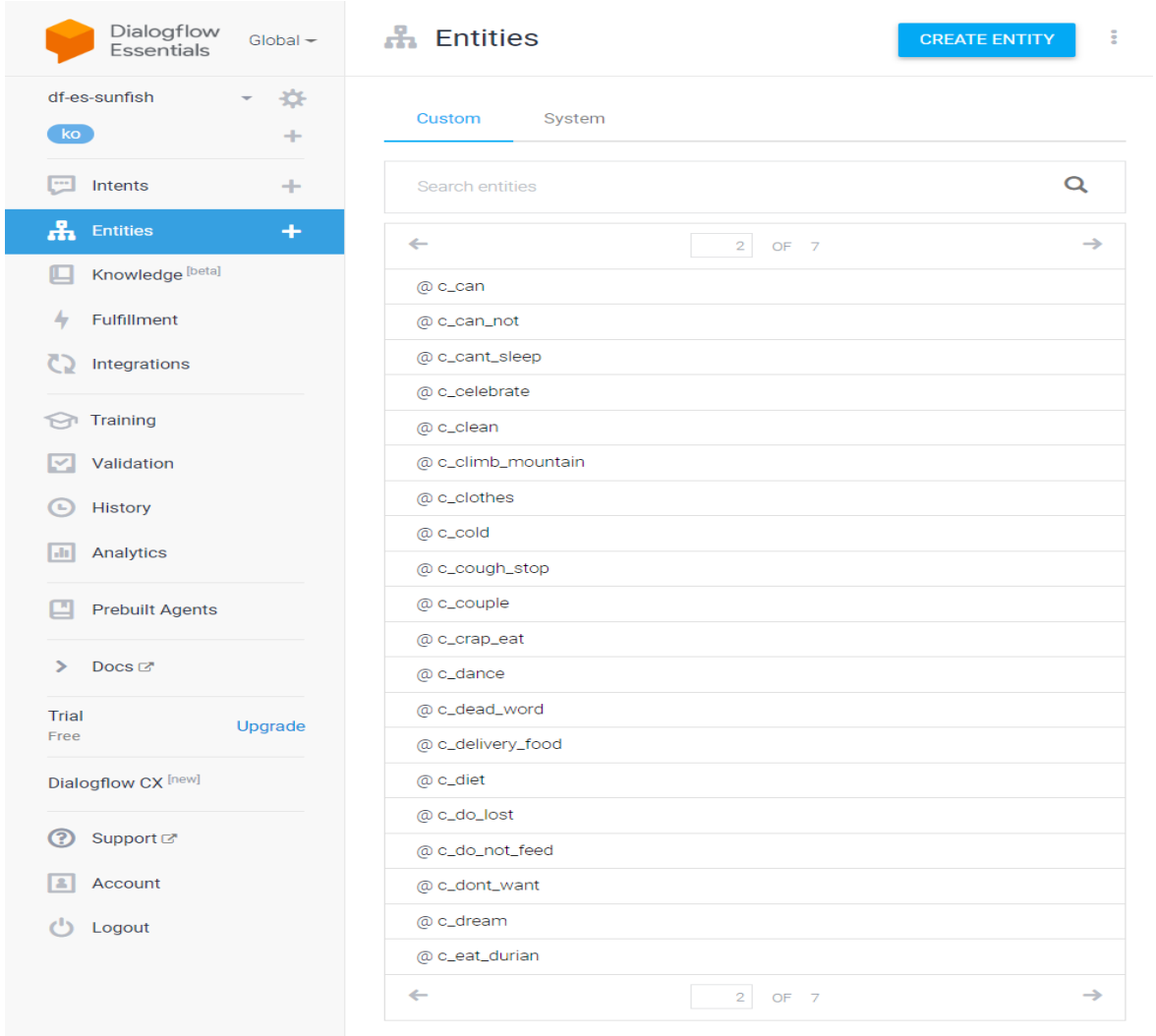
# 3

## Dialog-flow




(총 16 페이지)




(총 7 페이지)

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# 4

# Back-End

# Entity

## DfeCondition

```java
@Entity
@Getter
@NoArgsConstructor(access = AccessLevel.PROTECTED)
public class DfeCondition {
    no usages
    @ApiModelProperty(notes = "시퀀스")
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    1 usage
    @ApiModelProperty(notes = "df-entity 이름")
    @Column(nullable = false, length = 50)
    private String dfeName;

    1 usage
    @ApiModelProperty(notes = "df-entity 타입")
    @Column(nullable = false, length = 10)
    @Enumerated(EnumType.STRING)
    private DfeKind dfeKind;

    1 usage
    @ApiModelProperty(notes = "영향 여부")
    @Column(nullable = false)
    private Boolean isImpact;
```

## DieCondition

```java
@Entity
@Getter
@NoArgsConstructor(access = AccessLevel.PROTECTED)
public class DieCondition {
    no usages
    @ApiModelProperty(notes = "시퀀스")
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    1 usage
    @ApiModelProperty(notes = "intent 명")
    @Column(nullable = false, length = 150)
    private String intentName;

    1 usage
    private DieCondition(Builder builder) { this.intentName = builder.intentName; }

    1 usage
    public static class Builder implements CommonModelBuilder<DieCondition> {
        2 usages
        private final String intentName;

        no usages
        public Builder(DieConditionCreateRequest conditionCreateRequest) {
            this.intentName = conditionCreateRequest.getIntentName();
        }

        2 usages
        @Override
        public DieCondition build() { return new DieCondition( builder: this); }
    }
}
```

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# Enums

## DfeKind

```java
package shop.dev.choibk.api.game.logic.enums;

import lombok.AllArgsConstructor;
import lombok.Getter;

5 usages
@Getter
@AllArgsConstructor
public enum DfeKind {
    no usages
    POSITIVE( name: "긍정",  weighted: 10),
    no usages
    NEGATIVE( name: "부정",  weighted: -10),
    no usages
    NEUTRAL( name: "중립",  weighted: 0);

    no usages
    private final String name;
    no usages
    private final Integer weighted; // 곱셈 연산
}
```

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# Model

## DialogFlowResponse

```
@Getter
@Setter
public class DialogFlowResponse {
    no usages
    @ApiModelProperty(notes = "사람이 개복치에게 하는 말")
    private String queryText;

    no usages
    @ApiModelProperty(notes = "개복치가 사람에게 하는 말")
    private String fulfillmentText;

    no usages
    @ApiModelProperty(notes = "df- 의도 이름")
    private String intentName;

    no usages
    @ApiModelProperty(notes = "df- 사용된 Entity")
    private List<String> entitys;
}
```

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# Model

## DfeConditionCreateRequest

```java
@Entity
@Getter
@NoArgsConstructor(access = AccessLevel.PROTECTED)
public class DfeCondition {
    no usages
    @ApiModelProperty(notes = "시퀀스")
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    1 usage
    @ApiModelProperty(notes = "df-entity 이름")
    @Column(nullable = false, length = 50)
    private String dfeName;

    1 usage
    @ApiModelProperty(notes = "df-entity 타입")
    @Column(nullable = false, length = 10)
    @Enumerated(EnumType.STRING)
    private DfeKind dfeKind;

    1 usage
    @ApiModelProperty(notes = "영향 여부")
    @Column(nullable = false)
    private Boolean isImpact;
```

## DfeConditionCreateRequest

```java
@Entity
@Getter
@NoArgsConstructor(access = AccessLevel.PROTECTED)
public class DieCondition {
    no usages
    @ApiModelProperty(notes = "시퀀스")
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    1 usage
    @ApiModelProperty(notes = "intent 명")
    @Column(nullable = false, length = 150)
    private String intentName;

    1 usage
    private DieCondition(Builder builder) { this.intentName = builder.intentName; }

    1 usage
    public static class Builder implements CommonModelBuilder<DieCondition> {
        2 usages
        private final String intentName;

        no usages
        public Builder(DieConditionCreateRequest conditionCreateRequest) {
            this.intentName = conditionCreateRequest.getIntentName();
        }

        2 usages
        @Override
        public DieCondition build() { return new DieCondition( builder: this); }
    }
}
```

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# Model

## GameRequest

```
@Getter
@Setter
public class GameRequest {
    no usages
    @ApiModelProperty(notes = "개복치에게 요청할 말")
    @NotNull
    @Length(min = 5, max = 100)
    private String message;
}
```

## GameResponse

```
@Getter
@NoArgsConstructor(access = AccessLevel.PROTECTED)
public class GameResponse {
    1 usage
    @ApiModelProperty(notes = "사람 입력값")
    private String humanQueryText;

    1 usage
    @ApiModelProperty(notes = "ai 응답값")
    private String aiFulfillmentText;

    1 usage
    @ApiModelProperty(notes = "사망 여부")
    private Boolean isDead;

    1 usage
    @ApiModelProperty(notes = "변화되는 생명력 수치")
    private Integer changeVitalityFigures;
```

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# Repository

## DfeConditionRepository

```java
package shop.dev.choibk.api.game.logic.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import shop.dev.choibk.api.game.logic.entity.DfeCondition;

import java.util.Optional;

1 usage
public interface DfeConditionRepository extends JpaRepository<DfeCondition, Long> {
    no usages
    Optional<DfeCondition> findByDfeName(String dfeName);
}
```

## DieConditionRepository

```java
package shop.dev.choibk.api.game.logic.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import shop.dev.choibk.api.game.logic.entity.DieCondition;

1 usage
public interface DieConditionRepository extends JpaRepository<DieCondition, Long> {
    no usages
    long countByIntentName(String intentName);
}
```

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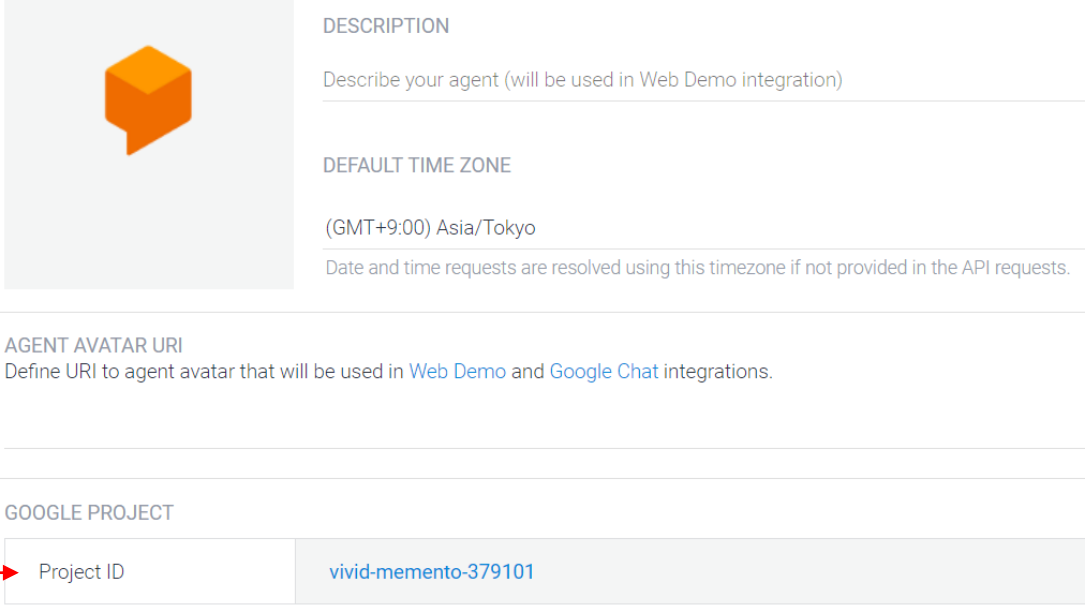
# Service

## DialogFlowService

```java
@Service
public class DialogFlowService {
    no usages
    public DialogFlowResponse detectIntentTexts(String callMsg) throws ApiException {
        String projectId = "vivid-memento-379101"; // ID 확인 필수

        String randomSessionId = UUID.randomUUID().toString();

        String langCode = "ko-KR";

        // Instantiates a client
        try (SessionsClient sessionsClient = SessionsClient.create()) {
            // Set the session name using the sessionId (UUID) and projectID (my-project-id)
            SessionName session = SessionName.of(projectId, randomSessionId);

            TextInput.Builder textInput = TextInput.newBuilder().setText(callMsg).setLanguageCode(langCode);

            // Build the query with the TextInput
            QueryInput queryInput = QueryInput.newBuilder().setText(textInput).build();

            // Performs the detect intent request
            DetectIntentResponse response = sessionsClient.detectIntent(session, queryInput);

            QueryResult result = response.getQueryResult();

            DialogFlowResponse resultResponse = new DialogFlowResponse();
            resultResponse.setQueryText(result.getQueryText());
            resultResponse.setFulfillmentText(result.getFulfillmentText());
            resultResponse.setIntentName(result.getIntent().getDisplayName());

            List<String> dfEntitys = new LinkedList<>();
            Struct temp1 = result.getParameters();
            Map<String, Value> temp2 = temp1.getFieldsMap();
            temp2.forEach((key, value) -> {
                dfEntitys.add(key);
            });
            resultResponse.setEntitys(dfEntitys);

            return resultResponse;
```

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# Service

## GameService (1)

```java
@Service
@RequiredArgsConstructor
public class GameService {
    1 usage
    private final DieConditionRepository dieConditionRepository;
    1 usage
    private final DfeConditionRepository dfeConditionRepository;
    1 usage
    private final DialogFlowService dialogFlowService;

    no usages
    public GameResponse doCalculation(GameRequest gameRequest) {
        DialogFlowResponse dialogFlowResponse = dialogFlowService.detectIntentTexts(gameRequest.getMessage());

        // --------- 즉사여부 확인 ----------
        // DieCondition 에서 위에서 가져온 displayName 에 해당하는 값이 있는지 확인함 -> count 로 확인
        long intentDuplicateCount = dieConditionRepository.countByIntentName(dialogFlowResponse.getIntentName());

        // 만약 데이터가 있으면 즉사처리
        if (intentDuplicateCount > 0) {
            return new GameResponse.Builder(dialogFlowResponse.getQueryText(),
                    dialogFlowResponse.getFulfillmentText(),
                    true, 0).build();
        } else { // 데이터가 없으면 점수 계산하는 2단계로 넘어감
            return new GameResponse.Builder(dialogFlowResponse.getQueryText(),
                    dialogFlowResponse.getFulfillmentText(),
                    false, calculateChangeVitalityFigures(dialogFlowResponse.getEntitys())).build();
        }
        // ---- 즉사여부 확인 끝 -----
    }
```

## GameService (2)

```java
    1 usage
    private int calculateChangeVitalityFigures(List<String> paramKeys) {
        List<DfeCondition> dfeConditions = new LinkedList<>();

        // 내가 가진 키들의 상세 데이터를 dfeConditions 에 모
        for (String key : paramKeys) {
            Optional<DfeCondition> tempResult = dfeConditionRepository.findByDfeName(key);
            tempResult.ifPresent(dfeConditions::add);
        }

        int weightValue = 0; // 최종 가중치
        int weightCount = 0; // 수치 영향 여부 총 갯수
        for (DfeCondition dfeCondition : dfeConditions) {
            weightValue += dfeCondition.getDfeKind().getWeighted();
            if (dfeCondition.getIsImpact()) {
                weightCount += 1;
            }
        }
        return weightValue * weightCount;
    }
}
```

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# Controller

## GameController

```
@Api(tags = "대화")
@RestController
@RequiredArgsConstructor
@RequestMapping("/v1/game")
public class GameController {
    1 usage
    private final GameService gameService;

    no usages
    @ApiOperation(value = "개복치와 대화하기")
    @PostMapping("/send")
    public SingleResult<GameResponse> sendMessage(@RequestBody @Valid GameRequest gameRequest) {
     return ResponseService.getSingleResult(gameService.doCalculation(gameRequest));
    }
}
```

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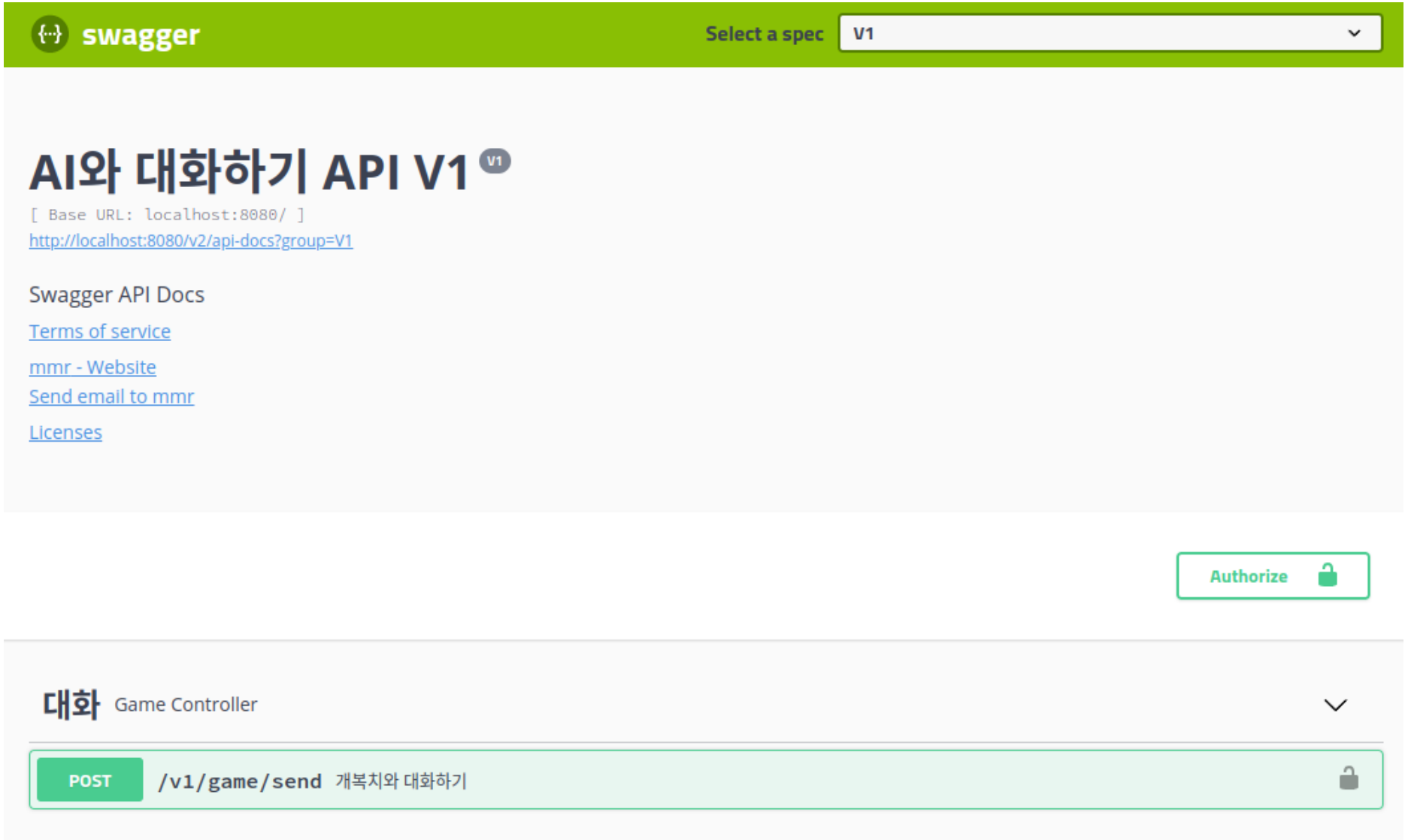
# Swagger-ui

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# Swagger-ui

## 대화 Game Controller

**POST** /v1/game/send 개복치와 대화하기

**Parameters**

| Name | Description |
|---|---|
| **gameRequest** * required<br>*(body)* | gameRequest<br>**Example Value** \| Model |

```
{
  "message": "밥 먹는다!"
}
```

사람이 입력할 대화

**Request URL**

```
http://localhost:8080/v1/game/send
```

**Server response**

| Code | Details |
|---|---|
| 200 | **Response body** |

```
{
  "isSuccess": true,
  "code": 0,
  "msg": "성공하였습니다.",
  "data": {
    "humanQueryText": "밥 먹는다!",
    "aiFulfillmentText": "맛있게 먹어~",
    "isDead": false,
    "changeVitalityFigures": 0
  }
}
```

AI가 출력할 대답

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# Swagger-ui

## 대화 Game Controller

**POST** /v1/game/send 개복치와 대화하기

**Parameters**

| Name | Description |
| --- | --- |
| **gameRequest** * required<br>*(body)* | gameRequest<br>**Example Value** \| Model |

```
{
  "message": "넌 멍청이야"
}
```

사람이 입력할 대화

**Request URL**

```
http://localhost:8080/v1/game/send
```

**Server response**

| Code | Details |
| --- | --- |
| 200 | **Response body** |

```
{
  "isSuccess": true,
  "code": 0,
  "msg": "성공하였습니다.",
  "data": {
    "humanQueryText": "넌 멍청이야",
    "aiFulfillmentText": "멍청이라니.. 너무해!!!!!",
    "isDead": false,
    "changeVitalityFigures": 0
  }
}
```

AI가 출력할 대답

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# Swagger-ui

## 대화 Game Controller

**POST** /v1/game/send 개복치와 대화하기

**Parameters**

| Name | Description |
| --- | --- |
| **gameRequest** * required<br>*(body)* | gameRequest<br>**Example Value** \| Model |

```
{
  "message": "야채 좋아하니?"
}
```

사람이 입력할 대화

**Request URL**

```
http://localhost:8080/v1/game/send
```

**Server response**

| Code | Details |
| --- | --- |
| 200 | **Response body** |

```
{
  "isSuccess": true,
  "code": 0,
  "msg": "성공하였습니다.",
  "data": {
    "humanQueryText": "야채 좋아하니?",
    "aiFulfillmentText": "당연히 좋아하지 ^^",
    "isDead": false,
    "changeVitalityFigures": 0
  }
}
```

AI가 출력할 대답

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# Swagger-ui

## 대화 Game Controller

**POST** /v1/game/send 개복치와 대화하기

**Parameters**

| Name | Description |
| --- | --- |
| **gameRequest** * required<br>*(body)* | gameRequest<br>**Example Value** \| Model |

```
{
  "message": "투수 오타니는 어떤사람 같니?"
}
```

↓

사람이 입력할 대화

**Request URL**

```
http://localhost:8080/v1/game/send
```

**Server response**

| Code | Details |
| --- | --- |
| 200 | **Response body** |

```
{
  "isSuccess": true,
  "code": 0,
  "msg": "성공하였습니다.",
  "data": {
    "humanQueryText": "투수 오타니는 어떤사람 같니?",
    "aiFulfillmentText": "멋있는 사람 같아",
    "isDead": false,
    "changeVitalityFigures": 0
  }
}
```

↓

AI가 출력할 대답

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# The End

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